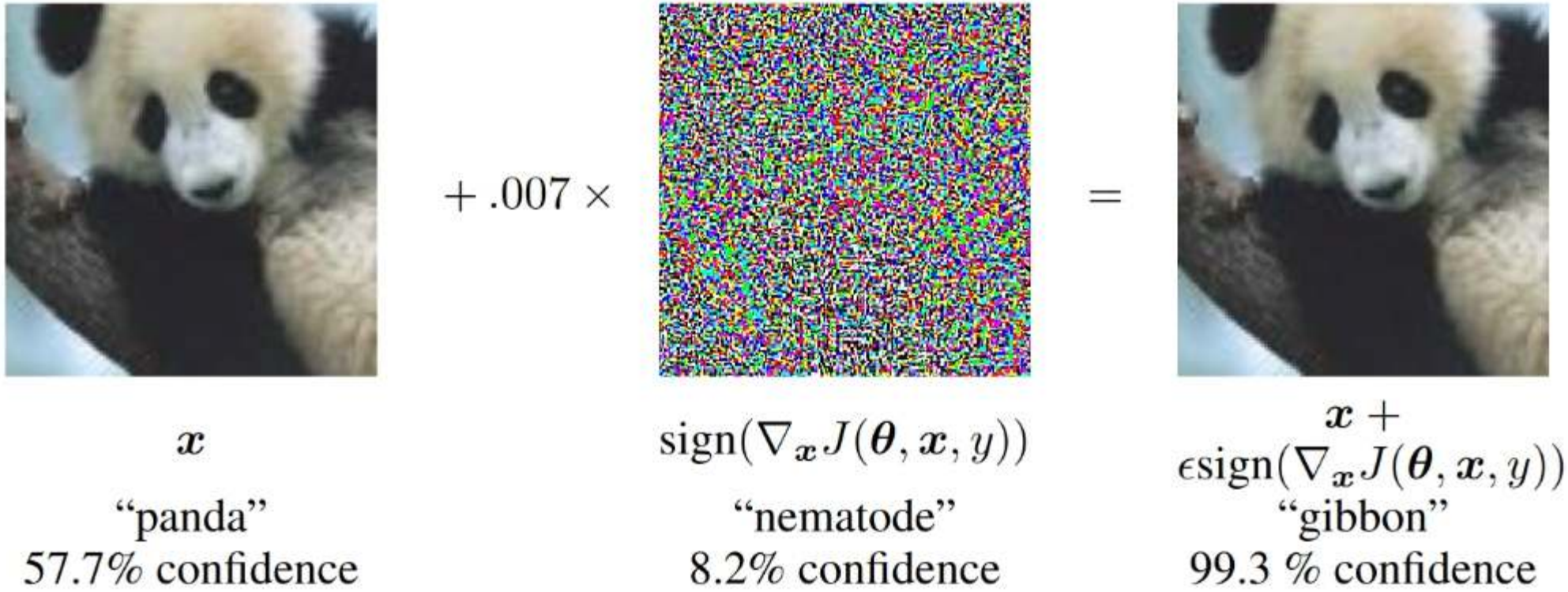
Crafting Adversarial Examples Using Linear Regions

Jules Soria, Julien Girard-Satabin, Romain Xu-Darme, Zakaria Chihani, Daniela Cancila

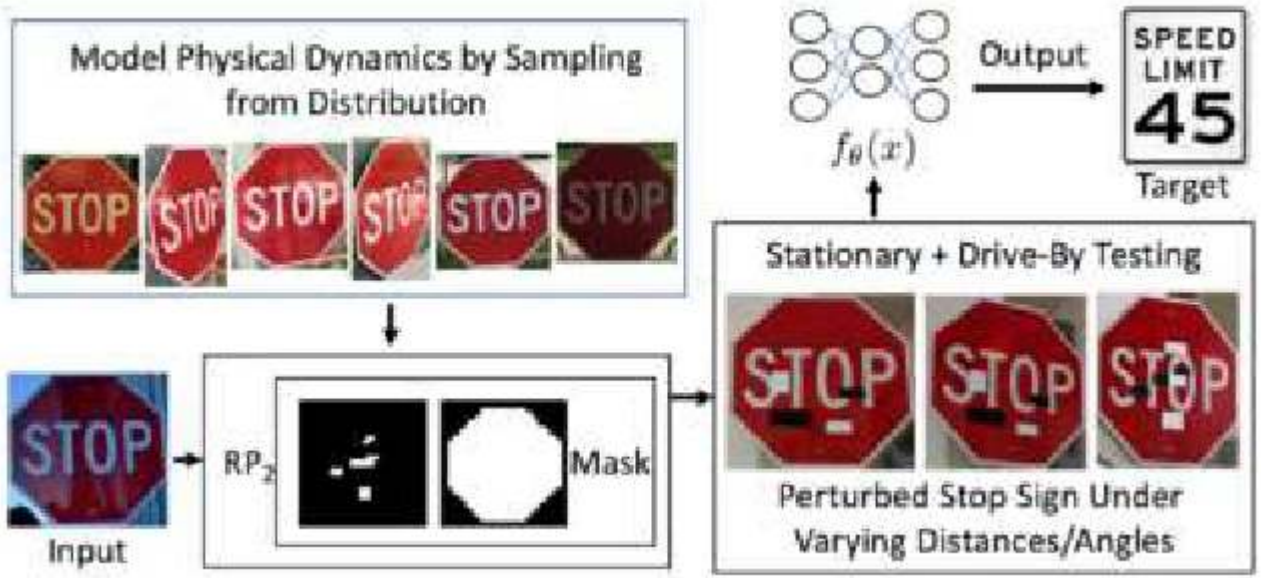
Université Paris-Saclay, CEA, List

AI models are not impervious to malfunctions



[1] Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and harnessing adversarial examples."

AI models' vulnerabilities can be exploited



[2] Eykholt, Kevin, et al. "Robust physical-world attacks on deep learning visual classification."

Overview of white-box adversarial attacks in ML

- FGSM [3] : $$x' = \begin{cases} x + \varepsilon \cdot sign(\nabla_x L(x, y)^T), untargeted \\ x - \varepsilon \cdot sign(\nabla_x L(x, t)^T), targeted\ on\ t. \end{cases}$$

- BIM [4], PGD [5] : $$\delta^{(k)} = \alpha \times sign\left(\nabla_x L\left(x^{(k)}, y\right)\right)$$

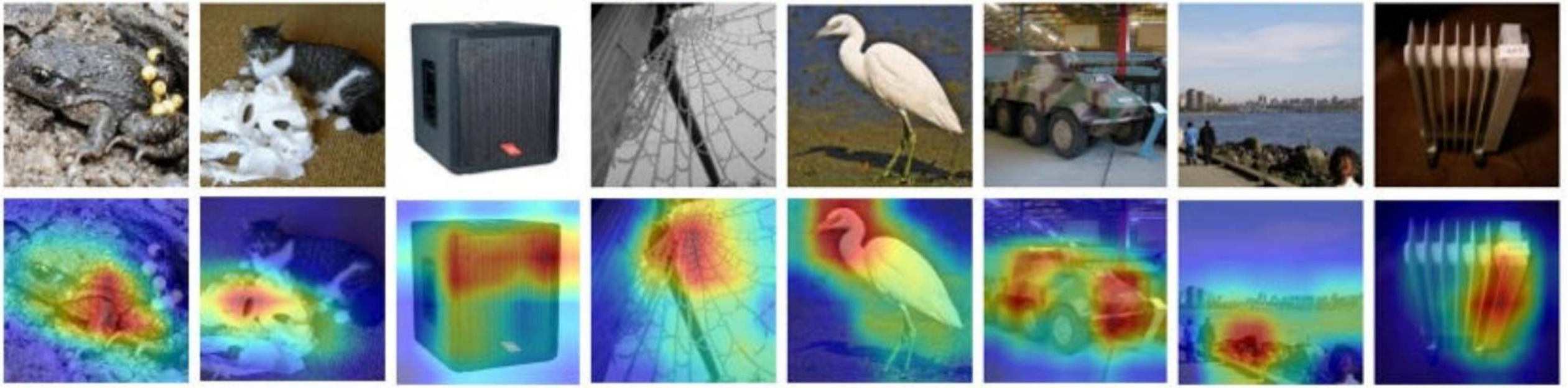
$$x^{(k+1)} = Clip_{x,\varepsilon}\left\{x^{(k)} + \delta^{(k)}\right\},$$

[3] Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy. "Explaining and harnessing adversarial examples."

[4] Kurakin, Alexey, Ian J. Goodfellow, and Samy Bengio. "Adversarial examples in the physical world."

[5] Madry, Aleksander, et al. "Towards deep learning models resistant to adversarial attacks."

Models can be explained



[11] Selvaraju, Ramprasaath R., et al. "Grad-cam: Visual explanations from deep networks via gradient-based localization."

Explanations can help detect adversarial attacks

Saliency maps give away what pixels are more important for the prediction :

→ they can help detecting adversarial images which force the model to « unnaturally » look at pixels that are not important



Original input:
zebra 99% confidence.



Explanation



Adversarial input:
assault rifle 15% confidence.

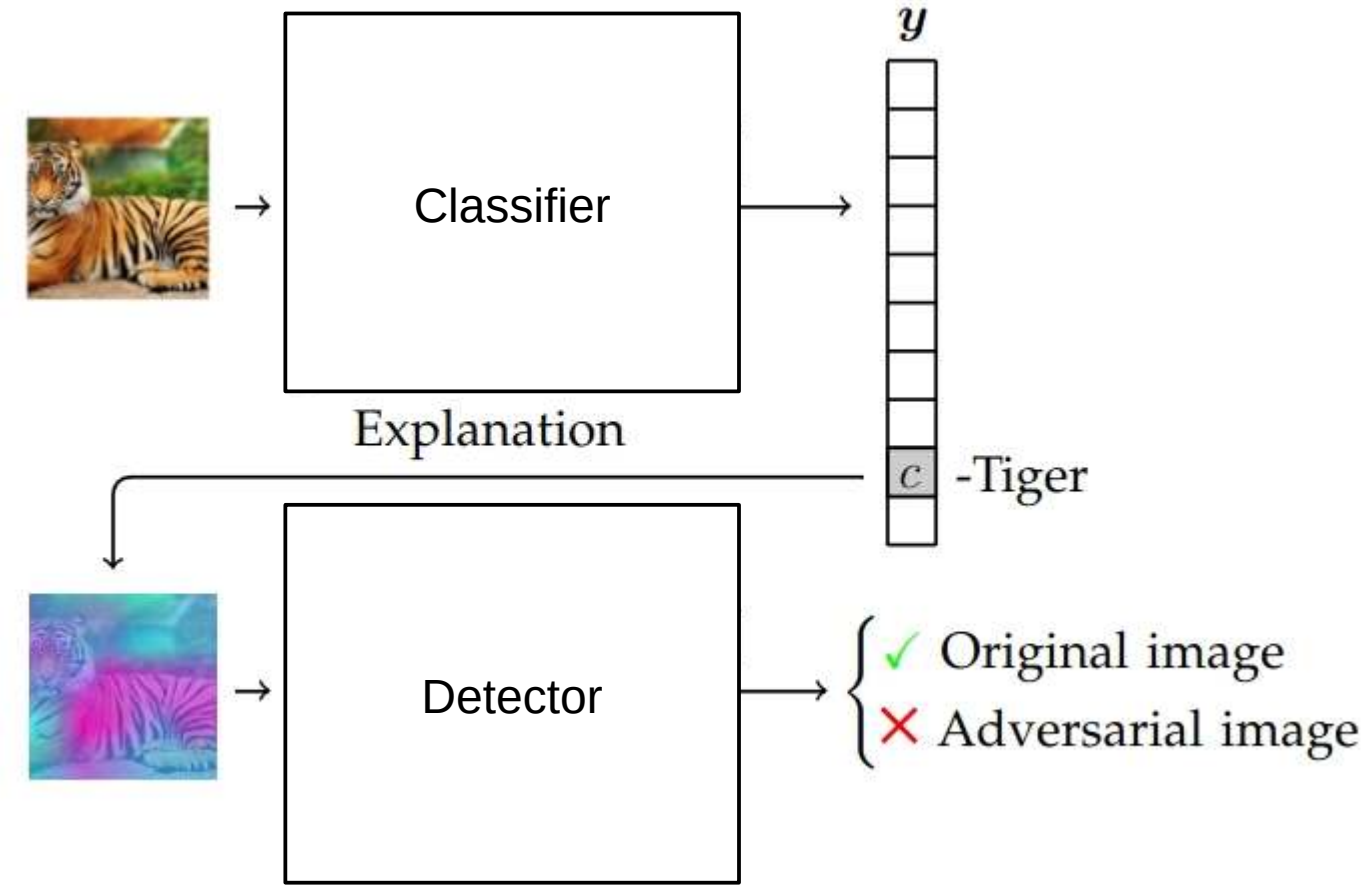


Explanation

[12] Stiff, Harald. "Explainable AI as a Defence Mechanism for Adversarial Examples."

[13] Zhang, Chiliang, Zhimou Yang, and Zuochang Ye. "Detecting Adversarial Perturbations with Saliency."

Explanations can help detect adversarial attacks



[12] Stiff, Harald. "Explainable AI as a Defence Mechanism for Adversarial Examples."

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An attack that bypasses XAI methods ?

- Main objective of the adversarial attack:
  - Predicted class changes: $argmax\, f(x) \neq argmax\, f(x')$
  - Small perturbation to the original sample: $|x - x'| < \epsilon$
  - Small perturbation to the saliency map: $|S_{argmax\, f(x)}(x) - S_{argmax\, f(x')}(x')| < \mu$

  → solve for $L(\hat{x}, x) = CE(f(\hat{x}), t) + MSE(\frac{df}{dx}\hat{x}, \frac{df}{dx}x)$ for target t ?

Preliminary results on CIFAR10

Same adversarial images produced as BIM[1] … → the gradient is not being backpropagated...

$$L(x^{(i)}, x) = CE(f(x^{(i)}), t) + MSE(\frac{df}{dx} x^{(i)}, \frac{df}{dx} x)$$

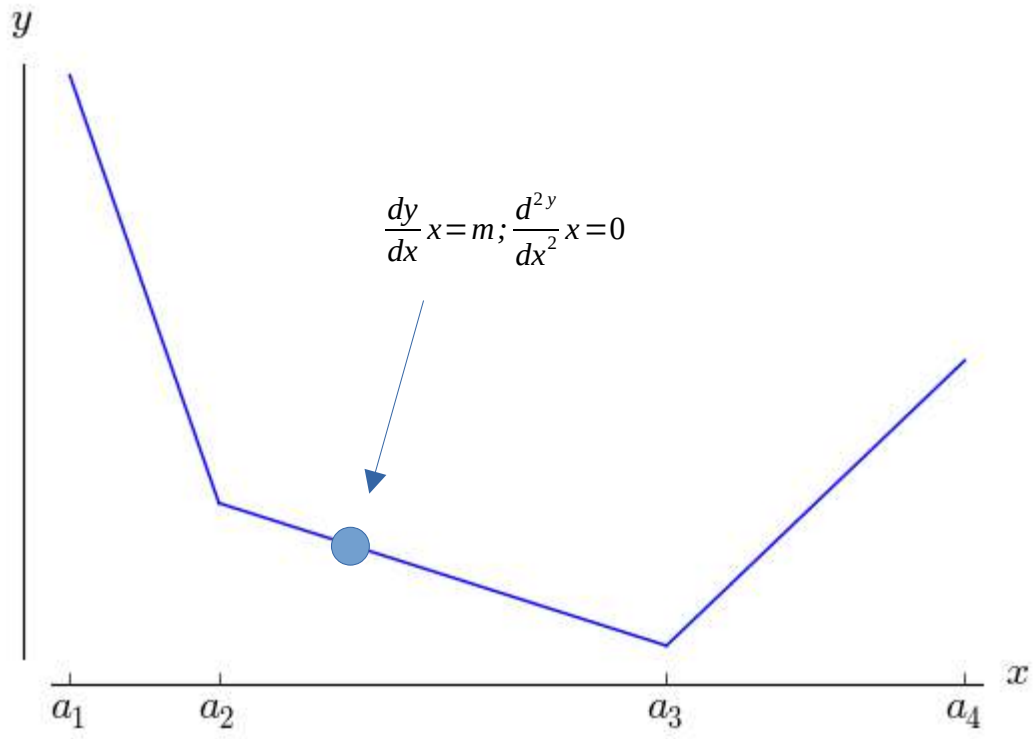
$$\delta^{(i)} = \alpha \times \text{sign}(\nabla L(\hat{x^{(i)}}, x))$$

$$x^{(i+1)} = \text{clip}_{x,\epsilon}(x^{(i)} + \delta^{(i)})$$

[1] for the same hyperparameters

Double back-propagation of gradients

… can only work if there are (nonzero) second-order derivatives in the model !



Another way to understand ReLU networks

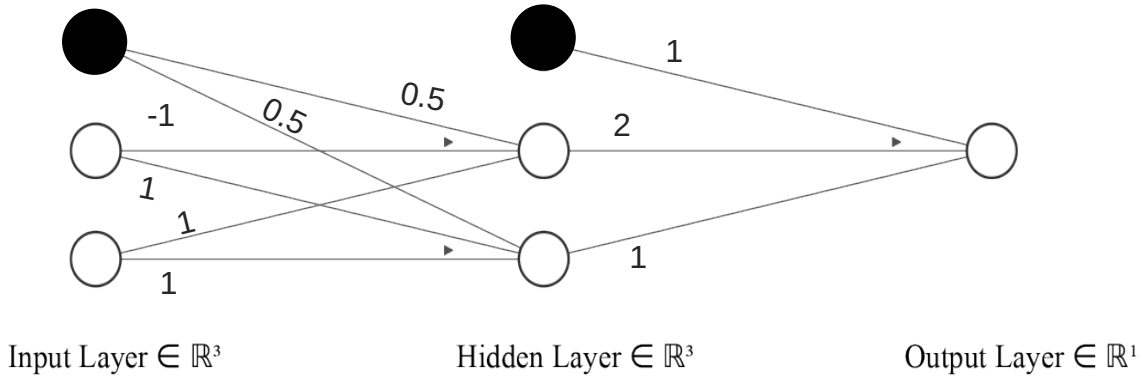


Fig 1. Small two-layer neural network with implicit ReLU activations

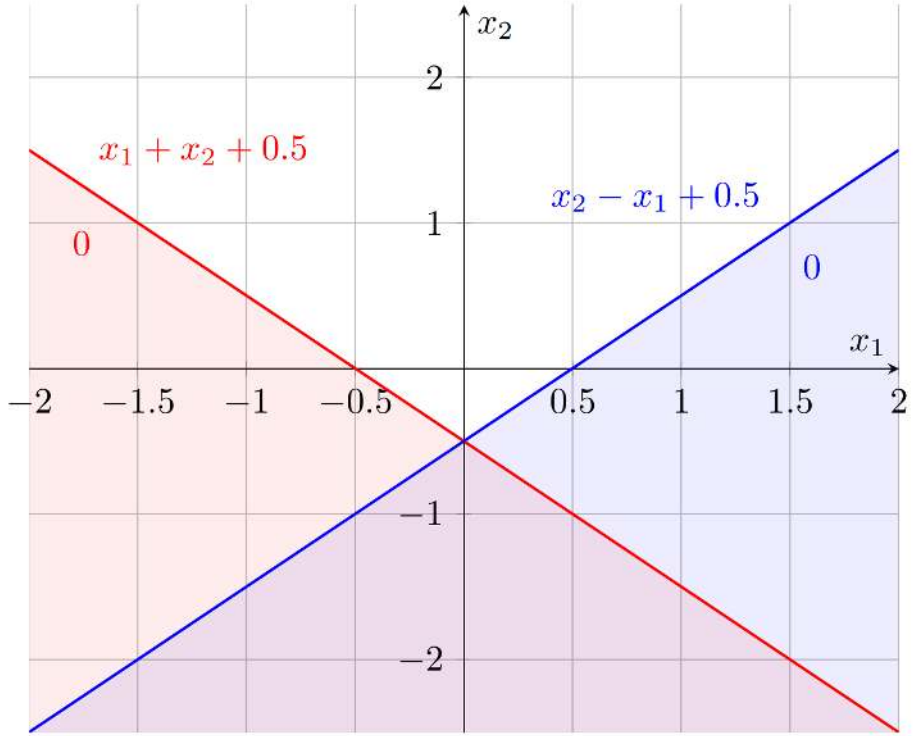


Fig 3. ReLU half-spaces between L_0 and L_1 in $\mathbb{R}^2$

[15] Girard-Satabin, Julien, et al. "DISCO Verification: Division of Input Space into COnvex polytopes for neural network verification."

Another way to understand ReLU networks

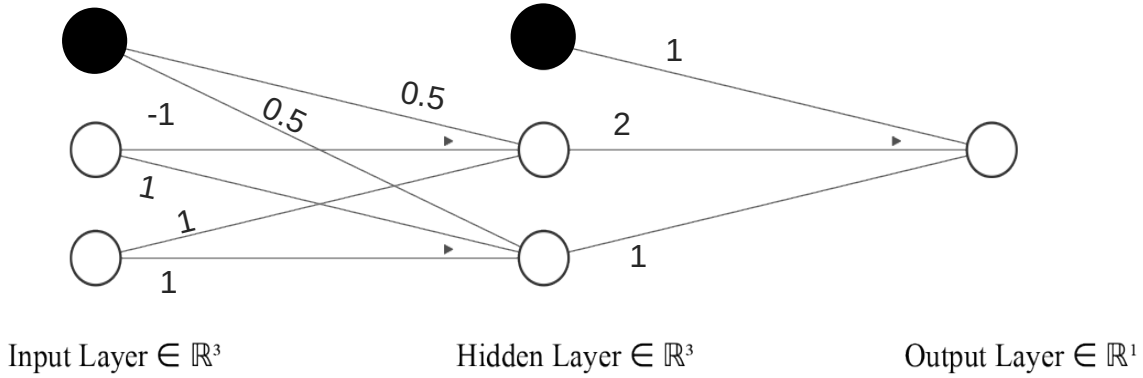


Fig 1. Small two-layer neural network with implicit ReLU activations

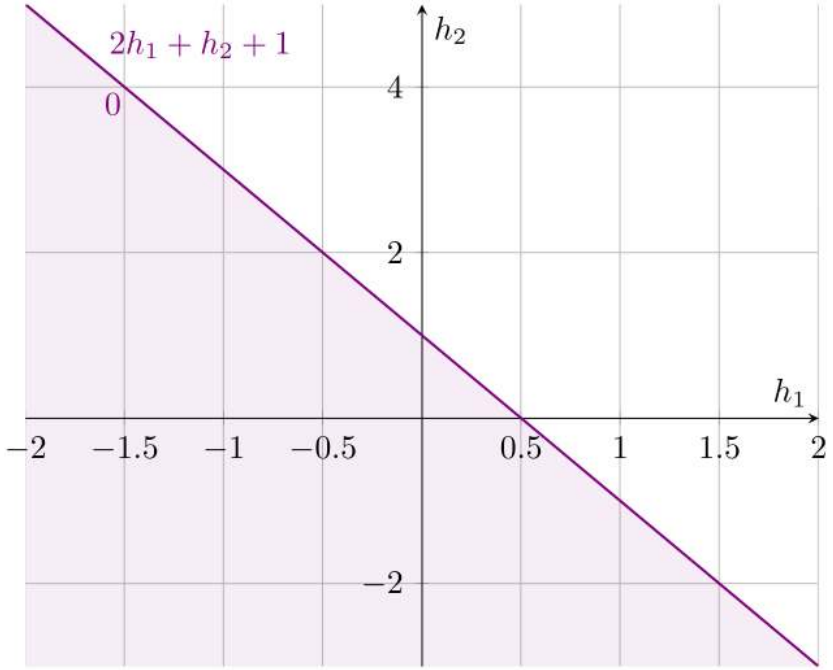


Fig 4. ReLU half-space between L_1 and L_2 in $\mathbb{R}^2$

[15] Girard-Satabin, Julien, et al. "DISCO Verification: Division of Input Space into COnvex polytopes for neural network verification."

Another way to understand ReLU networks

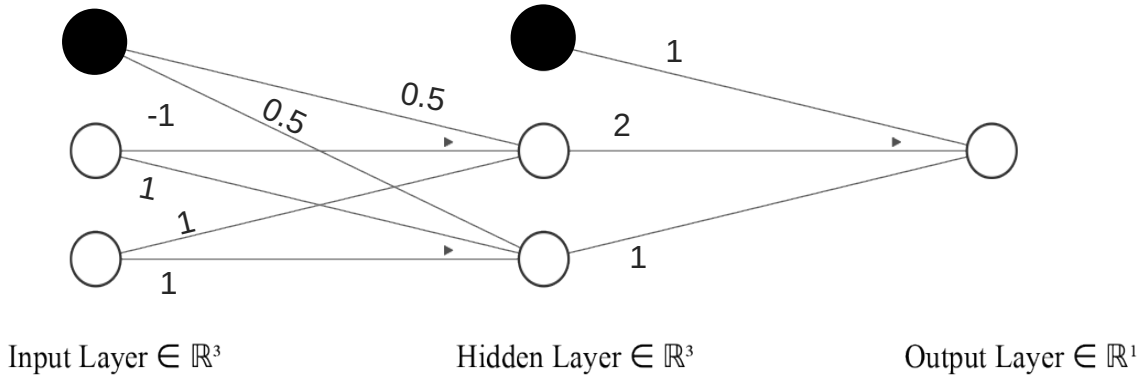


Fig 1. Small two-layer neural network with implicit ReLU activations

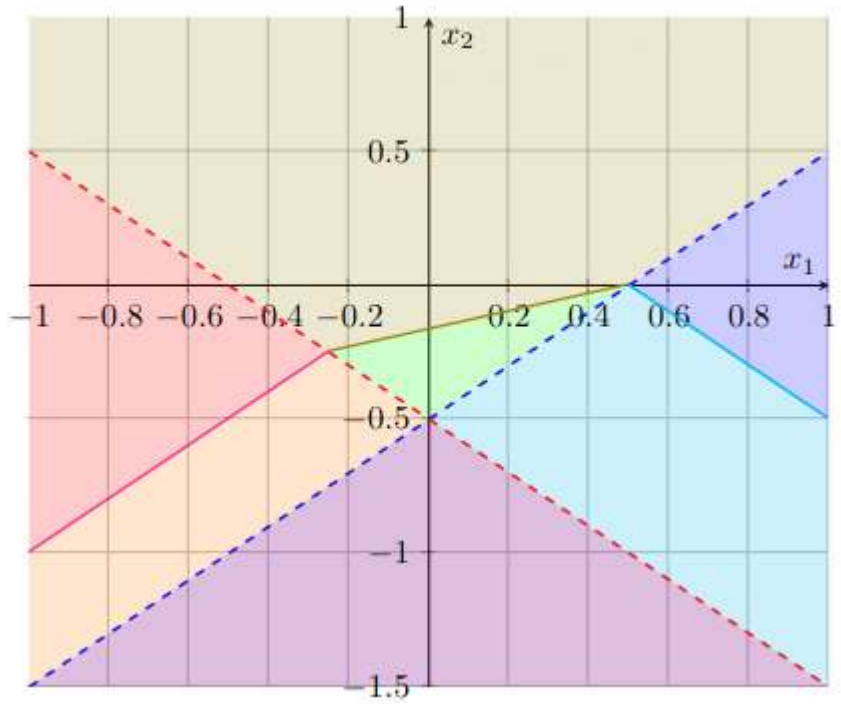


Fig 2. Associated linear regions from the network, in the input space $\mathbb{R}^2$

[15] Girard-Satabin, Julien, et al. "DISCO Verification: Division of Input Space into COnvex polytopes for neural network verification."

The same linear region, the same explanation

- With a forward pass, we can identify the linear region F of a sample image x, and « freeze » the ReLU activations of the neural network, such that it effectively behaves as an affine function :

$$f^F(x) = W^F \times x + b^F$$

- Two samples within the same linear region will have the same affine function attributed, and therefore the same gradients → the same explanation (for XAI methods that take gradient back-propagation as its paradigm)

Experimental Setup

- If the model is a deep ReLU network (i.e., no second-order derivatives) :
  - Calculate the gradient map for all classes
  - Find the class with the gradients closest to the predicted class (supposed true)
  - Generate adversarial samples with an attack method with this target class

- If the model is *not* a deep ReLU network (i.e., with second-order derivatives) :
  - Solve the optimization problem (that adds a constraint to the loss)

Preliminary results on CIFAR10 (Experiment n°1)

	Success Rate % (↑)	**Average Gradient MSE Loss (↓)**	**Average Cosine Similarity (↑)**
BIM (untargeted)	90,6 %	0,0067	-0,04
FGSM (untargeted)	67,7 %	0,0097	-0,01
Ours (w/ BIM) using cos_sim	58,66 %	0,0082	0,11
Ours (w/ FGSM) using cos_sim	18,22 %	0,0092	0,20

Preliminary results on CIFAR10 (Experiment n°1)

	Success Rate % (↑)	**Average Gradient MSE Loss (↓)**
BIM (untargeted)	90,6 %	0,0067
FGSM (untargeted)	67,7 %	0,0097
Ours (w/ BIM) using mse_loss	78,53 %	0,0067
Ours (w/ FGSM) using mse_loss	39,13 %	0,0075

Preliminary results on CIFAR10 (Experiment n°2)

		ReLU	ReLU + Sigmoid	ReLU + Tanh	SeLU	GeLu	Swish	Mish
	Accuracy	89,20 %	89,20 %	89,20 %	25,31 %	80,99 %	62,64 %	69,21 %
Success Rate (↑)	Ours	78,29 %	55,60 %	58,84 %	70,81 %	74,42 %	65,08 %	68,77 %
	BIM	78,29 %	55,62 %	58,91 %	70,66 %	74,49 %	65,06 %	68,82 %
Average MSE Loss (↓)	Ours	4,41	0,62	1,60	91,93	10,47	16,63	20,47
	BIM	4,41	0,62	1,60	94,95	12,37	18,08	23,4